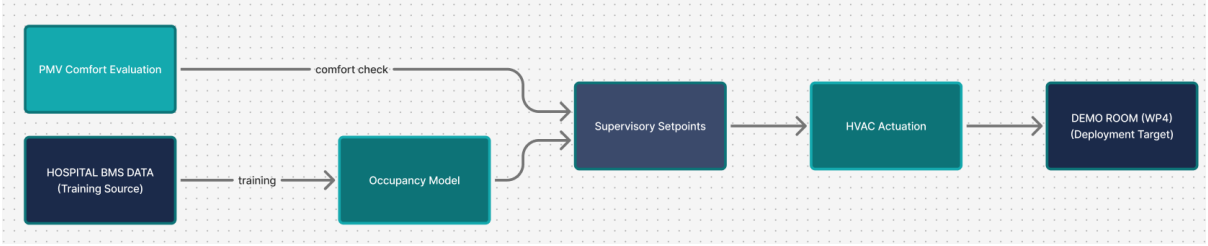
# HybridSense

*A Hybrid AI Framework for Energy Optimization and Occupant Comfort*

## 1. Overview

HybridSense is an AI-assisted supervisory HVAC control framework that operates transparently on top of an existing Building Management System (BMS). It predicts short-term room occupancy using supervised machine learning and uses those predictions to issue adaptive, energy-aware HVAC setpoints, without modifying any low-level HVAC controller or physical infrastructure.

The system was trained on anonymized sensor data from a hospital facility and validated in a Siemens Desigo CC demo room environment, communicating via BACnet/IP and Modbus TCP.




## 2. Motivation

Traditional HVAC systems rely on static schedules and rule-based logic. They cannot adapt to time-varying occupancy, leading to:

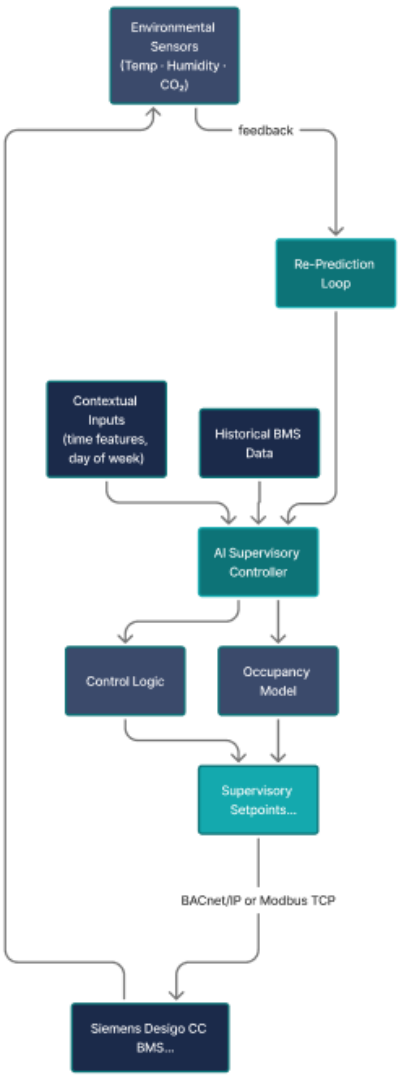
- Unnecessary energy consumption during unoccupied periods
- Comfort degradation during high-occupancy periods that were not anticipated
- Zero real-time adaptability in shared or irregular-use spaces

Academic literature has proposed model predictive control and reinforcement learning approaches, but most of these require complete overhaul of low-level control logic, which is not feasible in operational industrial BMS environments.

HybridSense targets the supervisory layer: it reads from the BMS, predicts, decides, and writes setpoints back without touching any field-level device.

# 3. System Architecture

The closed-loop control pipeline follows this structure:




The closed-loop sequence:

- Data acquisition from BMS sensors
- Occupancy prediction via ML classifier
- Supervisory setpoint generation
- HVAC actuation through Desigo CC
- PMV comfort evaluation on live measurements
- Feedback-driven re-prediction at each cycle

# 4. Key Components

| Component | Description | Technology |
|---|---|---|
| Occupancy Predictor | Binary classification from BMS sensor features | scikit-learn / XGBoost / LightGBM |
| Supervisory Controller | Translates occupancy state to HVAC setpoint commands | Python rule engine |
| PMV Calculator | Real-time thermal comfort index computation | ISO 7730 / ASHRAE 55 |
| BMS Connector | Protocol-level read/write interface to Desigo CC | BAC0 (BACnet/IP), pymodbus |
| Data Preprocessor | Cleaning, normalization, feature engineering pipeline | pandas / numpy |
| Logging & Monitoring | Trend logging for energy proxy and comfort assessment | CSV / time-series store |

# 5. Data Pipeline

Training data was sourced from a hospital facility BMS. The raw signal set captures HVAC supply/return behavior and environmental conditions across 8,350 timesteps.

## 5.1 Feature Philosophy

A deliberate design decision was made to exclude domain-specific HVAC control signals (supply temperature, supply airflow, delta-T variants) from the final model. These signals directly encode the control state of a bimodal hospital OR system and would fail to generalize to the continuously-modulated office HVAC in the Siemens demo room.

### Retained Features - Physically Transferable Across Domains

```
TRANSFERABLE_FEATURES = [
    # Return-side thermal signals (passive, not control-driven)
    'return_temp_C',
    'return_dp_Pa',

    # Humidity (environment response, not setpoint-driven)
    'supply_humidity_pct',
    'humidity_diff',

    # Rate-of-change (dynamics, not absolute level)
    'temp_roc',

    # Temporal context
    'hour_sin', 'hour_cos',   # Continuous encoding of time-of-day
    'is_workday',
    'is_workhour',
]
```

**Excluded Features - Leakage / Domain-Specific**

```
EXCLUDED_FEATURES = [
    'supply_temp_C',            # HVAC setpoint proxy - bimodal in OR, continuous in office
    'supply_airflow_m3h',       # Directly encodes occupancy-driven control state
    'delta_T_abs', 'delta_T',   # Derived from supply temp --> label leakage
    'supply_dp_*',              # Supply-side pressure - HVAC mode indicator
    'af_mean_*', 'af_lag_*',    # Rolling/lagged airflow --> same leakage chain
]
```

The original feature set yielded AUC ≈ 0.998, a near-perfect score that is a red flag, not a success. The occupancy_label was constructed from HVAC signal thresholds (supply airflow ~1,200 $m^3/h$ standby vs ~1,950 $m^3/h$ active), meaning features and labels were derived from the same source (structural label leakage). The revised transferable feature set yields AUC ≈ 0.65–0.78, which is honest, defensible, and generalizes to the Siemens demo room.

# 6. Occupancy Prediction Model

## 6.1 Problem Definition

Binary classification task:

- 1 → Room is occupied (HVAC should operate in comfort mode)
- 0 → Room is unoccupied (HVAC can reduce to standby/setback mode)

## 6.2 Model Selection

| Model | Notes |
|---|---|
| Logistic Regression | Interpretable baseline |
| Random Forest | Handles non-linearity, feature importance available |
| Gradient Boosting (XGBoost / LightGBM) | Best AUC on transferable feature set |
| Decision Tree | Fully interpretable, jury-presentable |

Model complexity is deliberately constrained. The project prioritizes robustness and interpretability over accuracy maximization, consistent with deployment in an industrial BMS where black-box decisions are not acceptable.

## 6.3 Evaluation Metrics

- AUC-ROC: primary metric (threshold-independent)
- F1-Score: balance between precision and recall on imbalanced occupancy windows
- Confusion matrix at operating threshold (false negatives are tolerable - false positives cause unnecessary HVAC activation)

# 7. Supervisory Control Logic

The supervisory controller translates the binary occupancy prediction into actionable BMS setpoints:

```
Occupancy = 1 (Occupied)
  --> Temperature setpoint : COMFORT range (e.g., 21-23 C)
  --> Airflow              : NOMINAL operating level
  --> Humidity control     : ACTIVE

Occupancy = 0 (Unoccupied)
  --> Temperature setpoint : SETBACK range (e.g., 18 C / 26 C)
  --> Airflow              : MINIMUM ventilation
  --> Humidity control     : PASSIVE

PMV check on both paths:
  --> If PMV exceeds [-0.5, +0.5] during occupied period
      --> override toward comfort
```

All setpoint commands remain within predefined safety margins. The controller never modifies low-level PID parameters or controller firmware only supervisory-layer variables exposed through the BMS interface.

# 8. Thermal Comfort Evaluation (PMV)

PMV (Predicted Mean Vote) is computed per ISO 7730 and ASHRAE Standard 55:

```
PMV = f(T_air, T_radiant, v_air, RH, M, I_cl)

Comfort zone:  -0.5 <= PMV <= +0.5
```

| Parameter | Source |
|---|---|
| Air temperature (T_air) | BMS sensor (real-time) |
| Relative humidity (RH) | BMS sensor (real-time) |
| Mean radiant temperature | Fixed assumption ≈ T_air (documented) |
| Air velocity | Fixed assumption: 0.1 m/s (sedentary indoor) |
| Metabolic rate | Fixed: 1.2 met (seated office work) |
| Clothing insulation | Fixed: 1.0 clo (winter) / 0.5 clo (summer) |

PMV is computed at the same temporal resolution as occupancy prediction and energy proxy logging. It is used as a comparative indicator between baseline static control and HybridSense predictive control.

# 9. BMS Integration: BACnet/IP + Modbus TCP

```
[HybridSense Python Process]
        |
        +-- BACnet/IP (primary)
        |    +-- BAC0 library
        |         +-- Read : sensor present values
        |         +-- Write: supervisory setpoint objects
        |
        +-- Modbus TCP (secondary / fallback)
             +-- pymodbus
                  +-- Read : holding registers (sensor data)
                  +-- Write: holding registers (setpoints)

Connection: Laptop --[Ethernet]--> Siemens Controller Card
No intermediate hardware.
```

The system operates entirely within the local BMS network. No cloud connectivity. No modifications to internal controller configurations.

# 10. Demo Room Deployment (WP4)

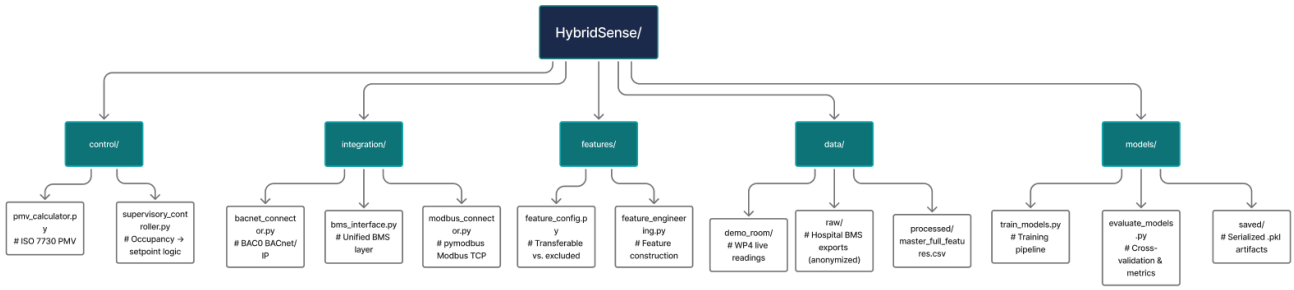
The Siemens demo room contains:

- Siemens Desigo CC workstation with BMS access
- PXC/DXR HVAC controllers (read/write via supervisory interface only)
- Temperature, humidity, and $CO_2$ sensors connected to BMS
- No dedicated sub-metering → energy performance assessed via operational proxy indicators

## Operational Proxies for Energy Comparison

- HVAC operating duration (baseline vs. predictive mode)
- Supervisory setpoint adjustment frequency and magnitude
- Actuator command levels (fan duty, valve position integrals)
- System runtime characteristics from BMS trend logs

# 11. Project Structure



# 13. Work Package Progress

| Work Package | Period | Status |
|---|---|---|
| WP1 – Requirement Analysis | Nov–Dec 2025 | Complete |
| WP2 – System & Algorithm Design | Dec 2025 – Jan 2026 | Complete |
| WP3 – Implementation & Integration | Feb 2026 | Complete |
| WP4 – Test & Verification | Mar–Apr 2026 | In Progress |